
Simulation of Wi-Fi Antennas Integrated into Glass

Comparative electromagnetic analysis of 2.4 GHz patch antennas

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Field: RF, antennas, and electromagnetic simulation
Primary tool: ANSYS HFSS
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Executive summary. Circular and rectangular patch antennas nominally designed for the 2.4 GHz Wi-Fi band were modeled in ANSYS HFSS. Three dielectric environments were compared for each geometry: FR4 in air, glass as the substrate in air, and FR4 integrated into a glass panel. The simulations reported resonant frequency, minimum S_{11} , maximum gain, and radiation patterns. The central finding is that glass cannot be treated as an electromagnetically neutral element: its role as either the substrate or the surrounding medium changes resonance, impedance matching, and gain. The results support retuning the antenna geometry for each integration scenario.

Engineering Problem

Discreetly integrating wireless connectivity into architectural surfaces requires an understanding of how the dielectric environment changes an antenna designed to operate near 2.4 GHz. This project used simulation to study the effect of glass on two patch-antenna geometries while maintaining a common comparison across three configurations.

Technical question: How do resonant frequency, impedance matching, and gain change when glass replaces FR4 or surrounds an antenna built on FR4?

Scope and Tools

Geometries	Circular and rectangular patches.
Configurations	FR4 in air; glass in air; FR4 integrated into glass.
Simulation environment	ANSYS HFSS, using coaxial-port excitation, open boundaries, and adaptive mesh refinement.
Reported sweep	2.3 GHz to 2.5 GHz in the original methodology. Some tabulated resonances fall outside this range and must be verified in the HFSS project files.
Metrics	Resonant frequency, minimum S_{11} , maximum gain, and radiation pattern.
Out of scope	Fabrication, physical measurements, durability, maintenance costs, and validation in a real installation.

Technical Workflow

1. Three-dimensional modeling of both geometries and material assignment.
2. Construction of three comparable dielectric environments for each geometry.
3. Definition of excitation, open boundaries, and adaptive meshing.
4. Simulation and extraction of S_{11} , gain, and radiation patterns.
5. Comparison of the six cases and interpretation of the effect of glass.

Reported Quantitative Results

The following values are transcribed from the result tables in the original report. They are simulation results, not experimental measurements.

Table 1: Reported simulation results for the circular patch.

Configuration	f_r (GHz)	Min. S_{11} (dB)	Max. gain (dBi)
FR4 in air	2.418	−29.11	3.07
Glass in air	2.179	−11.00	3.97
FR4 integrated into glass	2.139	−31.98	2.60

Table 2: Reported simulation results for the rectangular patch.

Configuration	f_r (GHz)	Min. S_{11} (dB)	Max. gain (dBi)
FR4 in air	2.490	−26.27	7.24
Glass in air	2.140	−7.70	3.25
FR4 integrated into glass	2.495	−29.40	6.66

Engineering Interpretation

- For the rectangular patch, replacing FR4 with glass shifted the reported resonance from 2.49 GHz to 2.14 GHz and reduced maximum gain from 7.24 dBi to 3.25 dBi.
- For the same geometry, the FR4-in-glass case reported 2.495 GHz, a minimum S_{11} of −29.40 dB, and 6.66 dBi; within the model, this case remained close to the baseline in frequency and gain.
- For the circular patch, both glass as the substrate and FR4 integrated into glass shifted the reported resonance below 2.2 GHz.
- The response was not uniform across geometries. The results therefore do not support a general claim that enclosing an antenna in glass will always preserve or always degrade performance.
- An integration targeting the 2.4 GHz band requires verification across the full target band and retuning of dimensions or materials when resonance shifts.

Visual Simulation Evidence

The following screenshots are retained as evidence of the results exported from HFSS. The software labels the polar plots as *Gain Plot*; therefore, they are not presented as independent directivity plots.

Circular Patch

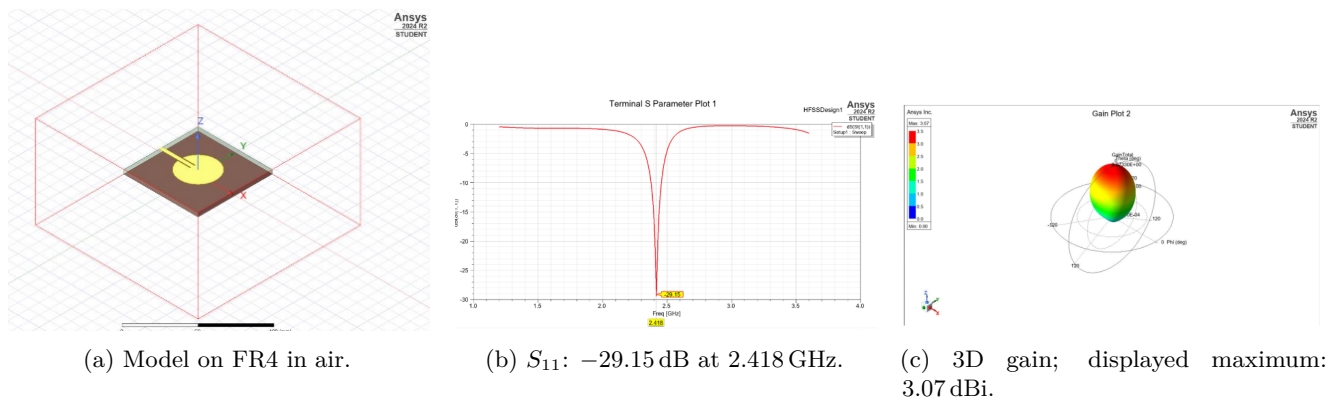
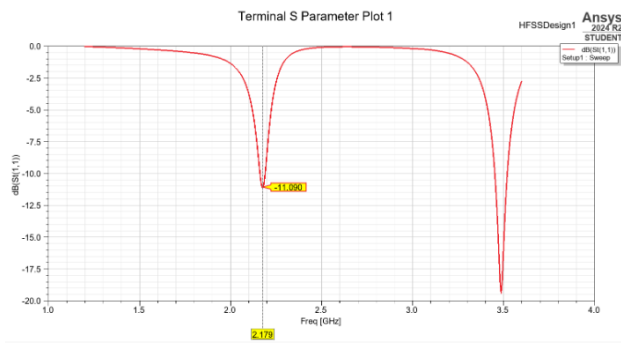
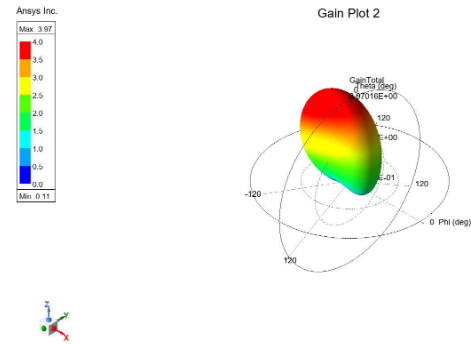


Figure 1: Circular patch: FR4-in-air configuration.

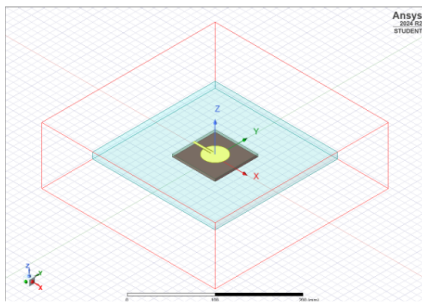


(a) S_{11} : -11.09 dB at 2.179 GHz.

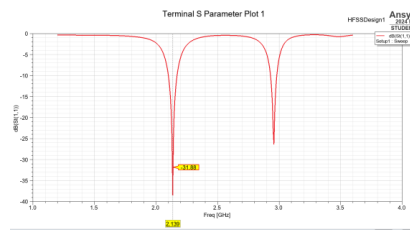


(b) 3D gain; displayed maximum: 3.97 dBi.

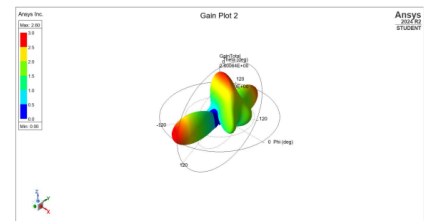
Figure 2: Circular patch: glass-substrate configuration.



(a) Model integrated into glass.



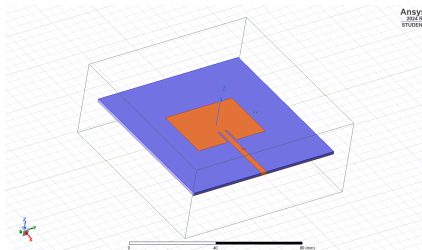
(b) S_{11} : -31.88 dB at 2.139 GHz.



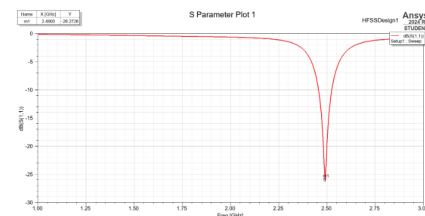
(c) 3D gain; displayed maximum: 2.60 dBi.

Figure 3: Circular patch: FR4 integrated into glass.

Rectangular Patch



(a) Model on FR4 in air.



(b) Series used in the summary: (c) 3D result with a displayed maximum of 7.24.

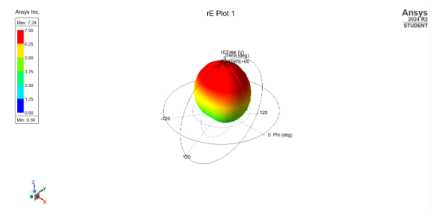
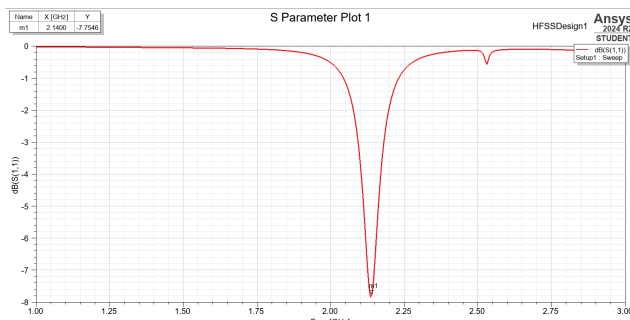
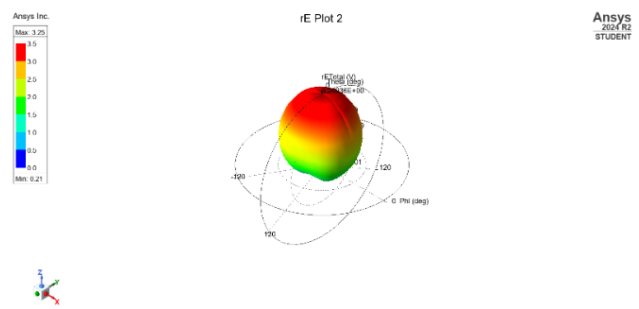


Figure 4: Rectangular patch: FR4-in-air configuration.



(a) S_{11} : -7.75 dB at 2.14 GHz.



(b) 3D result with a displayed maximum of 3.25.

Figure 5: Rectangular patch: glass-substrate configuration.

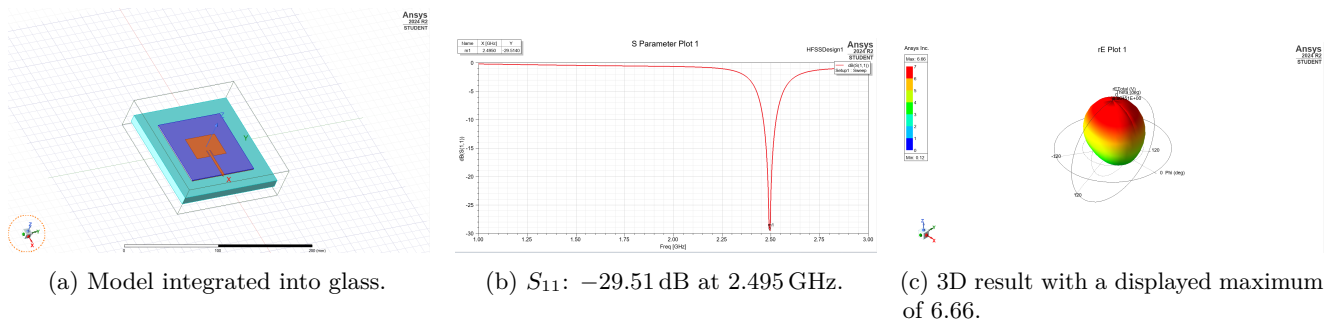


Figure 6: Rectangular patch: FR4 integrated into glass.

Conclusions

1. The dielectric environment appreciably changed resonant frequency, impedance matching, and gain in the simulated models.
2. Using glass as a direct substrate replacement produced reported resonances below the nominal 2.4 GHz band for both patches; a marked gain reduction was also observed for the rectangular patch.
3. Integrating FR4 into glass produced geometry-dependent responses: the rectangular patch remained close to the baseline, whereas the circular patch showed a substantial resonance shift.
4. The simulations provide evidence for a subsequent optimization stage, but do not independently demonstrate hardware performance or manufacturing feasibility.

Limitations and Next Iteration

- Confirm the electromagnetic properties and exact glass thickness in the HFSS project files.
- Repeat a sweep that includes all reported resonances and document the convergence criterion, mesh, and boundary conditions.
- Export S_{11} , gain, and directivity separately; the original document reuses some graphics for different metrics.
- Optimize at least one integrated geometry to recover operation at 2.4 GHz and compare its performance before and after optimization.
- Fabricate and measure a prototype before claiming physical feasibility, efficiency, durability, or performance in an architectural installation.

Technologies and Demonstrated Skills

Software	Ansys HFSS; 3D electromagnetic modeling.
RF and antennas	Microstrip patch antennas, S-parameters, resonance, gain, and radiation patterns.
Analysis	Controlled comparison of dielectric environments and quantitative interpretation of results.
Documentation	Technical synthesis, result traceability, and explicit identification of limitations and data requiring verification.